

Orthographic-to-Semantic Abstraction in Matched Spiking and Artificial Recurrent Networks: Depth, Settling, and Recurrent Dynamics

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Abstract

How recurrent neural systems transform elementary linguistic form into semantically organized activity remains poorly understood. We study a three-layer recurrent spiking neural network (SNN; 18,128 parameters) trained from one-hot grapheme events with a semantic triplet objective, without subword tokenization, pretrained semantic embeddings, attention, or a Transformer backbone. A parameter-matched artificial recurrent neural network (ANN) serves as a control. In a curated factorial diagnostic, the SNN shifts from orthographic to semantic organization across depth, with a rate-based abstraction index increasing from -0.854 in layer 1 to $+0.188$ in layer 3. Input order is critical: grapheme permutation reduces semantic triplet accuracy by 0.259, whereas removing precise output spike timing has little effect.

Extending the analysis beyond stimulus offset reveals that semantic organization continues to change during stimulus-free recurrent evolution. In the SNN, accumulated layer-3 abstraction increases from -0.221 at stimulus offset to $+0.153$ after eight settling steps and remains positive at $t + 72$ ($+0.163$, 95% CI $[0.037, 0.292]$). This effect is concentrated in the deepest layer. Causal interventions show that removing recurrent interaction during settling reduces this reorganization, while intrinsic neuronal dynamics retain a smaller contribution. Importantly, positive semantic organization is observed in the integrated population trajectory rather than in a stable instantaneous state.

The matched ANN also exhibits substantial post-stimulus reorganization, but with a slower temporal profile and point estimates approaching semantic neutrality at long aligned horizons. These results distinguish input temporal structure, recurrent temporal computation, and the temporal form of the final readout, suggesting that semantic organization in these small recurrent systems is better characterized as a trajectory-dependent dynamical process than as a static terminal representation.

1 Introduction

Spiking neural networks (SNNs) have recently scaled from spike-based document representations [Bialas et al., 2020]

to generative and general-purpose language models [Zhu et al., 2024, Xing et al., 2024, Tang, 2026]. Most work emphasizes architecture, task accuracy, or the prospect of sparse event-driven computation. We ask a complementary question: *how does linguistic organization change inside a small recurrent SNN that receives only elementary graphemic events?*

Direct processing of bytes or characters can support competitive language models and improve robustness on spelling-sensitive tasks [Xue et al., 2022]. It also creates a useful setting for representational analysis: semantic structure cannot enter through pretrained lexical embeddings. Our central questions are whether semantic organization becomes increasingly dominant across depth under semantic supervision, how it evolves after stimulus offset, what roles input order, recurrence, and spike timing play, and whether matched spiking and non-spiking systems exhibit the same temporal profile.

We make four contributions. First, a curated orthography $\times$ semantics diagnostic reveals a layerwise transition from form-dominated to meaning-dominated geometry under semantic supervision. Second, input perturbations and output readouts dissociate temporal *computation* from precise temporal output coding: sequence order is important, whereas exact final spike timing is not. Third, word-aligned settling analyses show that semantic separation continues to reorganize after stimulus offset and is expressed in the integrated population trajectory rather than as a stable instantaneous state. Fourth, causal interventions on post-stimulus dynamics show that recurrent interaction contributes directly to this reorganization, while matched SNN and ANN controls reveal distinct temporal profiles rather than a universally shared terminal geometry.

2 Related Work

SpikeGPT demonstrated sequential language generation with large backpropagation-trained SNNs [Zhu et al., 2024]; SpikeLM introduced elastic bi-spiking mechanisms for discriminative and generative language tasks [Xing et al., 2024]; and NeuronSpark reported end-to-end training of a 0.9B-parameter spiking language model from random initialization [Tang, 2026]. Earlier work learned low-dimensional spike-based document representations with

STDP [Białas et al., 2020], while other text classifiers first convert pretrained word embeddings into spike trains [Lv et al., 2023]. Our setting is intentionally smaller and diagnostic: semantic supervision is applied directly to recurrent encoders receiving one-hot graphemes.

Adaptive spiking neurons can expand the memory and computational capacity of recurrent SNNs [Bellec et al., 2018], and surrogate gradients enable optimization despite nondifferentiable spikes [Neftci et al., 2019]. To examine the resulting code, we use the van Rossum distance, whose time constant interpolates between sensitivity to temporal coincidence and firing-rate differences [van Rossum, 2001]. A recent MEG study also reported earlier orthographic than semantic information during natural reading [Wang et al., 2025]; this is an informative comparison, not a claim that our model is neurobiologically equivalent.

More broadly, recurrent computation need not terminate in a stable fixed-point representation. High-dimensional transient dynamics can provide fading memory and support readouts without stable internal states [Maass et al., 2002], while analyses of trained RNNs have related computation to trajectories and to fixed or slow regions of state space [Sussillo and Barak, 2013]. Recurrent connectivity can also generate robust, complex temporal trajectories after brief input [Laje and Buonomano, 2013]. These precedents motivate our distinction between trajectory-level organization and a terminal state, without implying that the present models instantiate biological neural dynamics.

3 Method

3.1 Models and input

The primary model is a three-layer recurrent SNN with adaptive leaky integrate-and-fire (ALIF) neurons. Decay parameters β are heterogeneous and learnable. Each grapheme is presented as a one-hot event; the model uses neither BPE nor pretrained semantic embeddings. A post-stimulus interval allows recurrent activity to evolve after the input ends. Training uses surrogate gradients, an activity/sparsity penalty, and a semantic triplet loss. All experiments ran on CPU.

The control ANN was designed to contain exactly 18,128 trainable parameters, matching the SNN. It preserves the same input, depth, width, recurrence, adaptation, decay initialization, seeds, data, and contrastive objective, while replacing binary spikes with adaptive leaky-tanh states. The comparison therefore isolates representational profile more closely than comparisons against a large pretrained language model, although neuron dynamics necessarily remain different.

3.2 Distances and evaluation

For anchor a , semantic positive p , and negative n , semantic triplet accuracy is

$$\text{STA} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[d(a_i, p_i) < d(a_i, n_i)]. \quad (1)$$

Spike trains are compared with van Rossum distance [van Rossum, 2001]; a rate-only readout discards within-window spike timing. Negatives are either random or *hard orthographic* distractors that resemble the anchor in form but not meaning.

To separate form and meaning, a curated diagnostic contains 48 pairs (12 per cell) in a 2×2 orthography $\times$ semantics design. At each layer l , predefined semantic and orthographic distance contrasts, S_l and O_l , yield

$$A_l = S_l - O_l, \quad (2)$$

where $A_l < 0$ denotes orthographic dominance and $A_l > 0$ semantic dominance. We additionally test relation-disjoint generalization, removing each evaluated positive relation in both directions from training, and supervised contextual transfer to eight new lexemes.

3.3 Temporal and dynamical controls

Output controls apply local or global spike-time shuffling while preserving each neuron’s spike count, as well as a rate-only readout. Input controls reverse or permute graphemes, insert random gaps, or replace order with a bag-of-graphemes representation. Timescale controls compare a single fixed β , fixed layerwise β , fixed heterogeneous β , and learnable heterogeneous β . Fixed-decay variants have 17,968 trainable parameters and the learnable variant 18,128, while all retain 18,128 state values. Principal reproducibility experiments use seeds 7, 19, 31, 43, and 59, a fixed set of 260 validation triplets, and repeated corruptions. Reported percentile-bootstrap intervals use seeds as the statistical unit; paired intervals are used for within-seed comparisons. Because lexical items are fixed, these intervals do not quantify uncertainty over alternative lexical samples.

3.4 Word-aligned settling and trajectory readouts

To distinguish terminal state from post-stimulus dynamical evolution, frozen checkpoints are re-evaluated with each sequence aligned to its final grapheme and extended with zero external input. Activity is analyzed at offsets $t + 0, +4, +8, +12, +20, +36$, and $+72$ relative to stimulus offset. This alignment removes the unequal amount of post-stimulus evolution produced by differences in word length under a fixed absolute simulation horizon.

We evaluate stimulus-only accumulated activity, post-stimulus accumulated activity, the full stimulus-plus-settling integrated trajectory, a local temporal window,

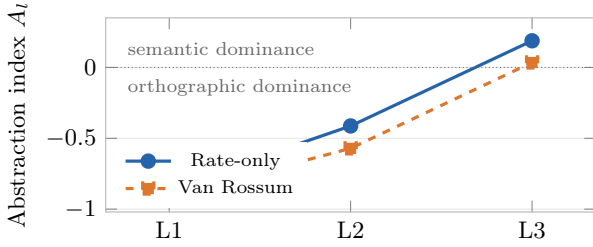


Figure 1: Fixed-horizon hierarchical diagnostic. Under the original absolute-horizon protocol, the orthography–semantics abstraction index crosses zero only in the deepest layer. The layer-3 rate-only estimate has a seed-level 95% CI of $[0.075, 0.304]$; lexical pairs are fixed.

and instantaneous population state. These readouts distinguish semantic organization distributed across the trajectory from organization present at any single time point.

To test the causal contribution of post-stimulus recurrence, each frozen checkpoint is additionally evaluated under four settling interventions. All interventions begin at the first time step after each sample’s final grapheme and are applied independently in every recurrent layer. Under *natural* settling, the frozen dynamics are left unchanged. In the *no-recurrence* condition, only the recurrent synaptic contribution, $W_{\text{rec}}h_{t-1}$, is set to zero; feed-forward/inter-layer drive and intrinsic leak, reset, and adaptation dynamics remain active. In the *intrinsic-only* condition, both feed-forward and recurrent synaptic drives are set to zero, leaving only state-dependent leak, SNN spike-reset, and adaptation dynamics. In the *reset* condition, the membrane or hidden state, previous output, and adaptation state are set to zero immediately before the first silent update, after which the unmodified natural dynamics resume. Training weights are unchanged in every condition.

4 Results

4.1 Depth reverses orthographic dominance

Within the curated factorial diagnostic, the rate-only abstraction index changes monotonically from $A_1 = -0.854$ to $A_2 = -0.413$ and $A_3 = +0.188$ (95% CI $[0.075, 0.304]$; Fig. 1). Van Rossum readouts show the same qualitative transition $(-0.911, -0.571, +0.035)$. At layer 3, graphically dissimilar synonyms are slightly closer on average than graphically similar semantic mismatches (0.627 vs. 0.661 van Rossum distance). This is an internal diagnostic on fixed lexical items, not a causal factorial experiment or evidence that the effect generalizes to the wider lexicon.

Triplet geometry corroborates the factorial result. Overall STA increases from 0.632 (L1) to 0.768 (L2) and 0.910 (L3), while the separation margin grows from 0.009 to 0.797. Hard orthographic negatives are initially closer than semantic positives, producing negative margins in L1 (-0.049) and L2 (-0.098) . L3 reverses the ordering (margin $= +0.156$, STA $= 0.640$). Depth thus expands the geometry while selectively correcting a form-based short-

cut; the result is a relative transformation rather than absolute convergence.

Post-stimulus activity also increases with depth (0.001, 0.021, and 0.045 for L1–L3). The word-aligned analyses below show that this activity is not merely residual persistence: in L3, semantic geometry continues to reorganize substantially after the external grapheme sequence has ended.

4.2 Post-stimulus evolution reorganizes semantic geometry

Aligning all sequences to their final grapheme reveals a substantial temporal transformation that was obscured by the original fixed absolute readout horizon. In the SNN, accumulated layer-3 abstraction is initially orthographically dominated at stimulus offset ($A_3 = -0.221$), becomes positive by $t + 4$ ($+0.069$), reaches $+0.153$ at $t + 8$ and $+0.185$ at $t + 12$, and remains reliably positive at $t + 72$ ($+0.163$, 95% CI $[0.037, 0.292]$; Fig. 2).

From $t + 0$ to $t + 8$, the paired increase in A_3 is $+0.375$ (95% CI $[0.344, 0.405]$). Over the same interval, overall STA increases by $+0.080$ $[0.042, 0.109]$, hard-negative STA by $+0.140$ $[0.070, 0.210]$, and the hard-negative separation margin by $+3.916$ $[2.754, 5.258]$.

The effect is layer-specific. The abstraction index changes little in L1 $(-0.894 \rightarrow -0.864$ at $t + 12$) or L2 $(-0.572 \rightarrow -0.460)$, whereas L3 changes from -0.221 to $+0.185$. Thus, post-stimulus transformation is concentrated in the deepest recurrent layer rather than being a uniform decay artifact throughout the network.

4.3 Semantic organization is expressed in the integrated trajectory

Positive layer-3 semantic dominance does not imply convergence to a stable semantic state. At $t + 72$, the SNN shows positive abstraction when the full stimulus-plus-settling trajectory is accumulated ($A_3 = +0.163$), but not in the instantaneous state (-0.495) , a local temporal window (-0.354) , or the post-stimulus interval considered alone (-0.299) .

Semantic organization is therefore distributed across the population trajectory generated during stimulus processing and subsequent recurrent evolution. Thus, “settling” should not be interpreted as the network eventually entering a semantic state; rather, post-stimulus evolution changes how the complete preceding trajectory separates semantic and orthographic relations. The results do not support an interpretation in terms of a stable semantic attractor or a terminal conceptual state.

4.4 Recurrence causally contributes to settling

Freezing the trained models and intervening only after stimulus offset separates recurrent interaction from passive persistence. In the SNN at $t + 12$, natural dynamics

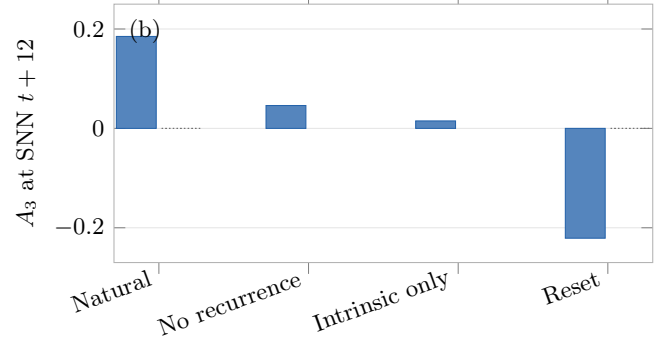
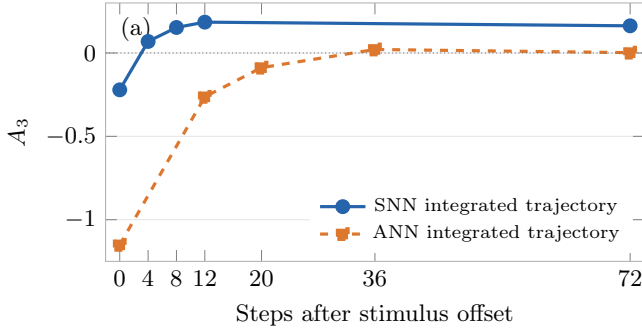


Figure 2: Word-aligned settling and causal interventions. (a) Integrated layer-3 abstraction reorganizes after stimulus offset. The SNN crosses into positive semantic dominance rapidly and remains positive at $t+72$ (95% CI [0.037, 0.292]), whereas the ANN changes more slowly and approaches neutrality at long offsets. Lines connect measured offsets only. (b) Intervening after the final grapheme shows that natural SNN recurrence contributes to the $t+12$ transformation; intrinsic dynamics retain a smaller contribution and reset restores the stimulus-offset value.

yield $A_3 = +0.185$, compared with $+0.046$ when recurrent drive is removed, $+0.015$ under the intrinsic-only condition, and -0.221 after full reset. Relative to the no-recurrence condition, natural recurrence increases A_3 by $+0.139$ (95% CI [0.090, 0.193]) and the hard-negative margin by $+2.086$ [1.231, 3.060]. The recurrent contribution remains detectable at $t+72$ ($\Delta A_3 = +0.131$, CI [0.069, 0.197]).

These interventions show that recurrent interaction contributes causally to post-stimulus semantic reorganization. However, improvement of the non-reset conditions relative to complete reset also indicates a contribution from intrinsic state dynamics. Recurrence therefore explains part, but not all, of the settling effect.

The ANN exhibits an even larger geometric change, but on a slower timescale. Its aligned A_3 changes from -1.151 at stimulus offset to -0.263 at $t+12$, -0.088 at $t+20$, $+0.021$ at $t+36$, and $+0.002$ at $t+72$. The $t+0 \rightarrow t+36$ change is $+1.173$ (95% CI [1.084, 1.243]). At $t+36$, natural ANN evolution yields $A_3 = +0.021$, compared with -0.467 without recurrent drive, a difference of $+0.488$ [0.361, 0.630]. Natural recurrence also improves STA by $+0.115$ [0.083, 0.148] and hard-negative STA by $+0.290$ [0.190, 0.350].

Thus, post-stimulus reorganization is not unique to spikes. Unlike the SNN, however, the aligned ANN point estimates approach neutrality at the longest evaluated offsets ($+0.021$ at $t+36$ and $+0.002$ at $t+72$). The original ANN $A_3 = +0.141$ was measured at a fixed absolute horizon, causing words of different lengths to receive different amounts of post-stimulus evolution. It remains a valid legacy measurement under that protocol, but it should not be interpreted as a word-aligned terminal-state comparison.

4.5 Temporal computation without precise temporal coding

Output-time perturbations provide little support for a precise temporal semantic code. Original van Rossum

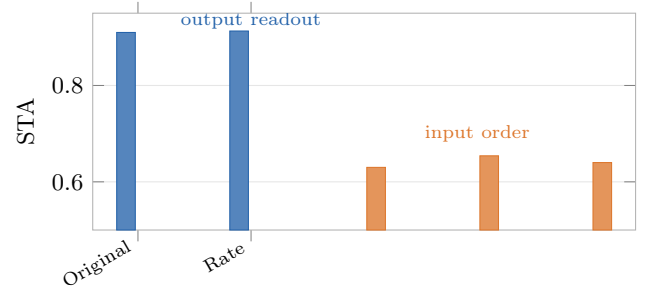


Figure 3: Temporal computation versus representation. Discarding output timing (rate-only) preserves STA, whereas disrupting input order produces large losses. Values use the corresponding primary readout: van Rossum for original and rate-only otherwise.

STA is 0.910 (95% CI [0.876, 0.935]); local shuffling is unchanged, global shuffling gives 0.886, and rate-only reaches 0.913 (95% CI [0.866, 0.945]). Similarly, the best single slow van Rossum scale ($\tau = 14$, STA = 0.914) is statistically indistinguishable from rate-only ($\Delta = 0.001$, CI [-0.012, 0.016]), while combining scales provides no benefit.

Input order, however, is essential (Fig. 3). Rate-only STA falls from 0.913 to 0.630 under reversal, 0.654 under permutation, and 0.640 for bag-of-graphemes input. Mild and strong gap jitter cause smaller drops to 0.867 and 0.824. On hard distractors, the positive L3 margin under correct order ($+0.156$) becomes negative under permutation (-0.109) or bagging (-0.129). The key distinction is therefore

ordered input $\rightarrow$ recurrent temporal evolution
 $\rightarrow$ integrated population trajectory,

not preservation of exact spike times as the semantic output format. Exact spike timing and temporal evolution are therefore distinct questions: the former contributes little to this readout, while the latter is central to the transformation.

Table 1: Matched 18,128-parameter model comparison using population/integrated-state metrics. The legacy fixed-horizon A_3 comparison is omitted because the word-aligned settling analysis supersedes its terminal-state interpretation. Bold marks the higher value; permutation drop is descriptive rather than “higher is better.”

Metric	SNN	ANN
IID STA	0.913	0.928
Hard orthographic STA	0.720	0.540
Permutation drop	0.259	0.133
Relation-disjoint	0.778	0.906
Contextual transfer	0.758	0.833

4.6 Heterogeneous neuronal dynamics

Under the van Rossum readout used for the decay ablation, learnable heterogeneous decays achieve overall STA = 0.910 and hard-negative STA = 0.640, compared with 0.892 and 0.490 for a single fixed β . The overall difference is inconclusive ($\Delta = 0.018$, CI $[-0.032, 0.065]$), but the hard-negative gain is reliable ($\Delta = 0.150$, CI $[0.020, 0.280]$), as is the L3 hard-margin gain (0.127, CI $[0.016, 0.226]$). The later population comparison uses rate-only SNN STA, explaining its higher hard-negative value of 0.720.

This supports heterogeneous dynamics, but not a strong claim for learning the timescales: fixed heterogeneity reaches hard STA = 0.520, and learned β values move only modestly. Mean $|\Delta\beta|$ decreases from 0.0127 in L1 to 0.0055 in L3, with no strong L3 correlation between β and post-stimulus or hard-example firing. The parsimonious interpretation is that initial heterogeneity supplies much of the benefit and learning makes small adjustments.

4.7 Matched SNN–ANN comparison

The matched ANN is numerically higher on IID STA (0.928 vs. 0.913), although the paired difference is inconclusive (-0.015 , CI $[-0.056, 0.012]$). It generalizes better on relation-disjoint evaluation (0.906 vs. 0.778; paired difference = -0.128 , CI $[-0.200, -0.056]$) and contextual transfer (0.833 vs. 0.758; paired difference = -0.075 , CI $[-0.142, -0.008]$; Table 1). Using population/integrated-state readouts, the SNN is better on hard orthographic negatives (0.720 vs. 0.540; difference = 0.180, CI $[0.040, 0.360]$) and shows a larger permutation drop (0.259 vs. 0.133).

The SNN spike rate is 0.090, whereas 98.6% of ANN states exceed 10^{-3} in magnitude. These quantities are not physically identical, but they establish a much sparser *operational state* in the SNN. They do not establish lower energy use, which would require a hardware- and operation-aware measurement.

4.8 Generalization and contextual transfer

When evaluated semantic relations are excluded from training in both directions, both models remain above chance: SNN STA is 0.778 (CI $[0.706, 0.844]$) and ANN STA is

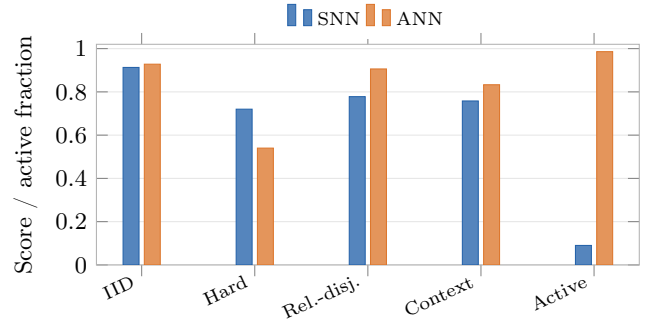


Figure 4: Distinct matched-model profiles. The ANN generalizes better, while the SNN better rejects deceptive orthographic negatives and has a lower active fraction. “Active” denotes spike rate for the SNN and $P(|\text{state}| > 10^{-3})$ for the ANN, so it is an operational, not energetic, comparison.

0.906 (CI $[0.850, 0.961]$). Thus the encoders do more than memorize the exact evaluated pairs, although the ANN generalizes substantially better.

For eight new lexemes, isolated-word STA begins near chance (SNN 0.487; ANN 0.546). After supervised exposure inside sentences, it rises to 0.758 for the SNN (gain = 0.271, CI $[0.208, 0.333]$) and 0.833 for the ANN (gain = 0.288, CI $[0.254, 0.313]$). This is supervised contextual transfer: contextual evidence can shape a representation later used in isolation, but the experiment does not demonstrate unsupervised semantic acquisition.

5 Discussion

The central result is a three-way dissociation between input temporal structure, recurrent temporal computation, and output temporal coding. Ordered grapheme sequences are required to construct the representation, yet precise spike timing within the population readout contributes little. At the same time, word-aligned analyses show that the representation is not static after the stimulus ends: semantic separation continues to reorganize during recurrent settling.

Crucially, the resulting semantic organization is not well described as a stable terminal population state. Positive semantic dominance is present in the full integrated trajectory while instantaneous, local-window, and post-only readouts remain orthographically dominated at long off-sets. The most conservative interpretation is therefore that semantic organization is expressed in the history of the recurrent population trajectory rather than in a single final state.

Causal settling interventions further show that recurrent interaction is not merely correlated with this transformation. Removing recurrent drive weakens semantic reorganization in both the SNN and ANN, while intrinsic state dynamics retain a smaller contribution in the SNN. Thus, post-stimulus organization reflects an interaction between recurrent communication and the internal dynamics of individual units.

The matched ANN demonstrates that post-stimulus se-

mantic reorganization is not unique to spikes. However, aligned temporal analysis revises the stronger interpretation of the original fixed-horizon comparison: the ANN shows a slower transition toward semantic neutrality, with long-horizon point estimates near zero rather than the reliably positive SNN estimate. The architectures therefore share recurrent reorganization but exhibit distinct temporal profiles. Outside settling, the SNN remains operationally sparser, more dependent on input order, and more robust to orthographic shortcuts; its relation-disjoint and contextual generalization are reliably lower than the ANN’s.

Heterogeneous neuronal timescales partially contribute to the SNN profile. Replacing them with one fixed decay mainly harms the difficult orthographic condition, while the global score changes little. Since fixed heterogeneity remains competitive and learned decays move only slightly, diversity of dynamics is better supported than strong timescale specialization.

Limitations. The dataset is small and Spanish-only, and the factorial diagnostic contains 12 pairs per cell. Its cells differ in semantic relation type and supervision exposure, lexical frequency is uncontrolled, and lexical selection may influence the contrasts. Seed-level confidence intervals do not represent uncertainty over alternative lexical samples. Semantic training and contextual transfer are supervised. Both networks are small and were evaluated on CPU. Sparse activity is not equivalent to energy efficiency, and no neuromorphic hardware measurement was made. The settling analyses characterize a trained dynamical system under zero external input; they do not demonstrate biological consolidation, deliberation, or attractor dynamics. Positive abstraction in the full trajectory should not be interpreted as a stable instantaneous semantic state. The original fixed-horizon factorial comparison also gave different words unequal post-stimulus durations; the word-aligned analysis corrects this confound. Finally, causal interventions identify contributions of recurrence within these trained models, not a unique mechanism of semantic cognition. The model receives graphemes rather than audio, and the findings do not establish speech understanding or biological plausibility. Broader corpora, preregistered lexical controls, additional matched recurrent baselines, and direct hardware measurements are needed.

6 Conclusion

A recurrent hierarchy receiving ordered elementary grapheme events can transform form-dominated activity into semantically organized population trajectories under semantic supervision. In the SNN, input order is important for constructing this organization, while precise output spike timing is not required to express semantic discrimination. Word-aligned analysis further shows that the transformation continues after stimulus offset: layer-3 semantic separation develops during stimulus-free recur-

rent evolution and depends causally, in part, on recurrent interaction.

The resulting organization should not be interpreted as a stable semantic terminal state. Positive semantic dominance is strongest when activity is considered as an integrated stimulus-plus-settling trajectory, whereas instantaneous and local post-stimulus states remain form-dominated. The matched ANN also undergoes substantial recurrent post-stimulus reorganization, but with a slower and quantitatively different temporal profile.

These findings motivate a view of representation in recurrent language systems as a temporally constructed dynamical trajectory rather than solely a static token-to-vector mapping. The next experiments test whether such dynamics can organize information across multiple temporal scales and eventually extend from elementary textual events to continuous acoustic input.

Code availability. Source code, experiment scripts, and reproducibility instructions are hosted at <https://github.com/Danval-003/snn-semantic-dynamics>.

References

- Guillaume Bellec, Darjan Salaj, Anand Subramoney, Robert Legenstein, and Wolfgang Maass. Long short-term memory and learning-to-learn in networks of spiking neurons. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Marcin Białas, Marcin Michał Mironczuk, and Jacek Mańdziuk. Biologically plausible learning of text representation with spiking neural networks. In *Parallel Problem Solving from Nature—PPSN XVI*, pages 433–447. Springer, 2020. doi: 10.1007/978-3-030-58112-1_30.
- Rodrigo Laje and Dean V. Buonomano. Robust timing and motor patterns by taming chaos in recurrent neural networks. *Nature Neuroscience*, 16(7):925–933, 2013. doi: 10.1038/nn.3405.
- Changze Lv, Jianhan Xu, and Xiaoqing Zheng. Spiking convolutional neural networks for text classification. In *International Conference on Learning Representations*, 2023.
- Wolfgang Maass, Thomas Natschläger, and Henry Markram. Real-time computing without stable states: A new framework for neural computation based on perturbations. *Neural Computation*, 14(11):2531–2560, 2002. doi: 10.1162/089976602760407955.
- Emre O. Neftci, Hesham Mostafa, and Friedemann Zenke. Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks. *IEEE Signal Processing Magazine*, 36(6):51–63, 2019. doi: 10.1109/MSP.2019.2931595.

- David Sussillo and Omri Barak. Opening the black box: Low-dimensional dynamics in high-dimensional recurrent neural networks. *Neural Computation*, 25(3):626–649, 2013. doi: 10.1162/NECO_a_00409.
- Zhengzheng Tang. NeuronSpark: A spiking neural network language model with selective state space dynamics. *arXiv preprint arXiv:2603.16148*, 2026.
- M. C. W. van Rossum. A novel spike distance. *Neural Computation*, 13(4):751–763, 2001. doi: 10.1162/089976601300014321.
- Lijuan Wang, Steven Frisson, Yali Pan, and Ole Jensen. Fast hierarchical processing of orthographic and semantic parafoveal information during natural reading. *Nature Communications*, 16:8893, 2025. doi: 10.1038/s41467-025-63916-y.
- Xingrun Xing, Zheng Zhang, Ziyi Ni, Shitao Xiao, Yiming Ju, Siqi Fan, Yequan Wang, Jiajun Zhang, and Guoqi Li. SpikeLM: Towards general spike-driven language modeling via elastic bi-spiking mechanisms. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 54698–54714. PMLR, 2024.
- Linting Xue, Aditya Barua, Noah Constant, Rami Al-Rfou, Sharan Narang, Mihir Kale, Adam Roberts, and Colin Raffel. ByT5: Towards a token-free future with pre-trained byte-to-byte models. *Transactions of the Association for Computational Linguistics*, 10:291–306, 2022. doi: 10.1162/tacl_a_00461.
- Rui-Jie Zhu, Qihang Zhao, Guoqi Li, and Jason K. Eshraghian. SpikeGPT: Generative pre-trained language model with spiking neural networks. *Transactions on Machine Learning Research*, 2024.